AMIT YADAV

Backend Developer

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PROFESSIONAL SUMMARY

Backend Developer and B.Tech graduate (2026) trained in Software Development & Applied AI at Masai School. Built and deployed backend applications using Python, FastAPI, SQLAlchemy ORM, PostgreSQL, and JavaScript. Strong foundation in DSA (30+ LeetCode problems), modular REST APIs, JWT and owner-based authorization, automated pytest, Git workflows, and LLM API integration.

PROJECTS

ZOMATO NOTES – Internal Knowledge Base for On-Call Support Engineers | Live Demo

GitHub Repository | Python, FastAPI, JavaScript, Supabase/PostgreSQL, Groq API

- Built and deployed a FastAPI knowledge base for on-call engineers to log incidents and find past fixes: JWT-protected CRUD APIs, keyword search, and tag filtering over an indexed Supabase (PostgreSQL) schema; typical endpoints respond in 3-9 ms.
- Designed normalized PostgreSQL users and notes tables with SQLAlchemy ORM and foreign-key constraints.
- Structured the backend into separate layers (routes, database operations, models, schemas, authentication, search, AI services) so each can be changed and tested independently.
- Implemented owner-based authorization: editing or deleting a note requires a matching x-user-id, and deletion also requires a token; verified with 401/403/200 tests.
- Implemented linear search (O(n)), iterative and recursive binary search (O(log n)) for exact-title lookup on a database-sorted list, and insertion sort ($O(n^2)$) to rank results by keyword score.
- Wrote 36 pytest integration tests (auth, CRUD, search, filtering, validation, errors, permissions); GitHub Actions runs them on every push and pull request to gate Render deploys.
- Added semantic search with sentence-transformers and cosine similarity; lazy-loaded the model for a 4.6 s startup and ~385ms warm queries.
- Added X-Process-Time middleware, Python error logging (visible in Render), and a /ping health endpoint for monitoring.
- Integrated the Groq API to generate note content, suggest tags from titles, and summarize notes.
- Tested live by Masai mentors; acted on their code-review feedback by restricting CORS from ten origins to one documented frontend origin and adding an inline "Get AI Summary" action to every note card.

MY_LLM – AI Chat Assistant | Live Demo

GitHub Repository | Python, FastAPI, JavaScript, Supabase/PostgreSQL, Groq API

- Built and deployed an AI chat app with FastAPI and JavaScript, storing messages and conversation IDs in Supabase (PostgreSQL) for persistent history.
- Called the Groq API server-side through FastAPI and streamed responses to the frontend using Server-Sent Events (SSE).
- Applied Pydantic request validation, tested the API with Postman and Swagger UI, and wrote a README for setup and usage.
- Configured automatic deployments from GitHub to GitHub Pages (frontend) and Render (backend) on every push.

LINKSNAP – URL Shortener | Live Demo

GitHub Repository | Python, FastAPI, JavaScript, PostgreSQL/Supabase, Docker, Render

- Built and deployed a URL shortener that converts long URLs into short links and records every click with a timestamp.
- Implemented atomic click recording (INSERT + counter UPDATE in one transaction); verified rollback with a monkeypatched pytest failure (18 tests passing).
- Dockerised with docker-compose (API + PostgreSQL); CI via GitHub Actions; deployed on Render with Supabase.
- Used a Git feature-branch workflow: built the frontend on a separate branch and merged it via GitHub pull request.

SKILLS

- **Languages & DSA:** Python, JavaScript, SQL, DSA(30+ LeetCode Problems: Arrays, Dictionaries, Binary/Linear Search, Sorting)
- **Backend:** FastAPI, REST APIs, SQLAlchemy, JWT Authentication, Owner-Based Authorization, Pydantic validation, middleware, error handling
- **Frontend:** HTML5, CSS3, Vanilla JavaScript, DOM Manipulation, Async Fetch API, LocalStorage, responsive UI
- **Databases:** PostgreSQL (Supabase), Relational Schema Design, Normalization, Table Relationships, Indexing, Foreign Keys
- **AI:** LLM API integration (Groq), embedding-based semantic search, AI summarization
- **DevOps and Testing:** Pytest (Unit & Integration Testing), FastAPI TestClient, HTTPX, Git & GitHub (Feature Branches), GitHub Actions (CI/CD), Docker, Postman, Swagger UI, Render, GitHub Pages

EDUCATION

Software Development, Masai School — March 2026 – July 2026 | Remote

B.Tech, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV) — November 2022 – April 2026 | Bhopal

PROFESSIONAL EXPERIENCE

Web Development Intern, InternPe — March 2024 – July 2024 | Remote

- Built a responsive e-commerce website with HTML, CSS, and JavaScript, backed by a FastAPI REST API and PostgreSQL database serving category-wise product listings.
- Implemented shopping-cart functionality with quantity updates and automatic total-price recalculation, persisting cart data in PostgreSQL through FastAPI endpoints (add, update, remove), with validation to block negative quantities.